

AI for Financial Practitioners

Lecture 1 · Part II

Autonomy, Delegation and System Governance

A pedagogical monograph

How institutions can delegate discretion to intelligent systems without confusing capability with authority or automation with accountable autonomy

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ABSTRACT

This monograph treats autonomy as delegated discretion rather than as an intrinsic property of a model. It offers a ladder of bounded delegation, from informational assistance to conditionally authorized action, and identifies the evidence required before authority can expand. Governance is presented as executable architecture: identity, permissions, limits, confirmations, audit records, exception paths and re-vocation must accompany capability. Through financial examples, the monograph shows how an institution can gain the speed and scale of agentic systems while retaining accountable ownership of objectives, constraints and consequential decisions.

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LECTURE 2 · PART II

Autonomy, Delegation and System Governance

A visual guide to the key ideas of the monograph

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1 CAPABILITY IS NOT AUTHORITY
A system may be technically capable, but it is only authorized to act within a defined mandate.

Technical Capability	Institutional Authority
What the system can do	What the system is permitted to do
Access information	Defined objectives
Analyze and reason	Scope of permitted actions
Generate content	Conditions and constraints
Make calculations	Escalation requirements
Use tools and take action	Human oversight
	Audit and accountability

Capability is a property of the technology.
Authority is a decision of the institution.

2 AUTONOMY IS DELEGATED DISCRETION
Autonomy is not intelligence. It is the freedom to decide or execute actions without seeking approval at every step.

The objective is not to delegate the most, but to delegate the right amount under the right conditions, with the right safeguards.

3 DELEGATION SHOULD BE GRADUATED AND BOUNDED
Different tasks require different levels of autonomy, with clear boundaries.

Domain	Typical Tasks	Appropriate Autonomy	Key Boundaries
Research and analysis	Gather information, analyze data, draft reports	Recommendation or supervised execution	Approved sources, no external disclosure
Portfolio analytics	Scenario analysis, risk measurement	Supervised execution	Approved models and data, validated assumptions
Client interaction	Prepare materials, answer standard queries	Assistance or recommendation	Approved content, escalate complex or sensitive cases
Trading execution	Execute within mandate, monitor positions	Supervised or autonomous (with strict limits)	Position limits, risk checks, real-time monitoring
Corporate actions	Process routine actions, reconcile data	Supervised execution	Defined scope, validation and audit trail

4 GOVERNANCE MUST BECOME ARCHITECTURAL
Permissions, evidence, monitoring and intervention should be designed into the system from the outset.

Governance is not an afterthought. It is part of the architecture.

5 FROM DELEGATION TO ACCOUNTABILITY
Every level of autonomy requires evidence, oversight and a clear allocation of responsibility.

Define and Authorize	Operate and Monitor	Evaluate and Learn	Intervene and Adapt
Set objectives Define scope Specify constraints Assign authority Document approval	Execute or assist Log actions Monitor performance Detect anomalies Maintain audit trail	Assess outcomes Compare to objectives Identify failures Refine procedures Update authority if appropriate	Escalate when needed Pause or revoke authority Correct and recover Update safeguards Reassess risk and scope

Delegation is only sustainable when it is connected to continuing oversight, evaluation and the credible possibility of intervention.

6 THE DESTINATION: GOVERNABLE AUTONOMY
The sophisticated institution is not the one that delegates the most, but the one that delegates well.

A more capable, trustworthy and economically valuable institution.

"Autonomy is not the absence of control, but the product of better control."

Part I organized intelligence into tools, skills, memory, agents and coordinated responsibilities. Part II asks the question that follows immediately: what authority should that organization receive? Autonomy is not a property that mysteriously appears when a model becomes sufficiently intelligent. It is a decision by an institution to delegate discretion.

Introduction: the moment the assistant is allowed to act

MAIN IDEA

The transition from assistance to action changes the institutional problem even when the underlying model remains exactly the same.

The research service is now genuinely useful. It can retrieve approved documents, calculate controlled metrics, compare new information with institutional memory, prepare a draft and ask a reviewer to challenge its conclusions. Analysts have used it for several months. The natural next request sounds almost trivial: if the service already knows when an important event has occurred, why should it wait for an analyst to send the alert?

That question marks a profound transition. Before the change, the system produced information for a human being. After the change, it can alter the institutional environment. It may send a message that changes a portfolio manager's attention, update a record on which future analysis depends, open a case, request information from another department or perhaps initiate a financial action. The underlying model may be exactly the same. What has changed is authority.

This distinction is essential because contemporary discussion often treats autonomy as the natural consequence of intelligence. A model becomes more capable, we connect it to tools, and eventually we describe the result as autonomous. Institutions cannot afford that conceptual shortcut. Technical capability answers the question *can the system do this?* Delegation answers a different question: *may the system do this without obtaining approval at this moment?*

Human organizations understand the distinction intuitively. A talented analyst may be capable of executing a trade, but the institution does not infer trading authority from talent. A lawyer may understand a contract but not possess authority to bind the company. A risk officer may identify a limit breach but not be authorized to change the limit. Capability and authority are deliberately separated because consequences matter.

The same discipline should govern artificial systems. Part II therefore treats autonomy

as an institutional design problem. We will ask what is being delegated, how far discretion extends, what evidence justifies the delegation, what controls remain active while the system operates and how authority can be withdrawn when conditions change. The objective is not maximum autonomy. The objective is governable autonomy.

1 Capability is not authority

MAIN IDEA

A system's ability to perform an action says nothing about whether the institution has legitimately permitted that action.

Suppose a language model can draft a convincing investment recommendation. Nothing in that capability tells us whether it should be permitted to distribute the recommendation to clients. Suppose an agent can call a trading API. Nothing about the existence of the connection tells us which securities it may trade, in what quantities, for which accounts or under whose mandate. The action space of the software and the authority of the institution are separate objects.

This sounds obvious when stated plainly, yet digital systems can blur the distinction because permissions are often embedded in technical configurations. If an agent receives credentials that allow a database to be modified, the technical possibility may quietly become operational authority. A prototype built for convenience can therefore acquire powers that were never deliberately delegated.

A mature architecture makes authority explicit. For each action class, the institution should know whether the system may only propose the action, prepare it for approval, execute it after specific approval or execute it within predefined limits without contemporaneous approval. These are materially different arrangements even if the analytical reasoning preceding the action is identical.

The distinction also protects us from an unhelpful hierarchy in which more autonomy is assumed to be more advanced. Consider two systems. The first uses a highly capable model to analyze a complex restructuring but produces only a draft for an experienced committee. The second is a simple deterministic service that automatically sweeps excess cash between approved accounts at the end of each day. The second may possess more operational authority even though it exhibits less cognitive sophistication. Autonomy is therefore not a ranking of intelligence.

This matters in finance because the consequences of action are asymmetric. An erroneous paragraph in a draft can be corrected. An erroneous external communication can create legal and reputational consequences. An erroneous payment can move

money. An erroneous trade can create market exposure before anyone has time to read an explanation. As consequences increase, the evidence required for delegation should generally become more demanding.

Authority should also attach to a specific object. Permission to read a filing does not imply permission to contact the issuer. Permission to update a draft does not imply permission to overwrite the approved research record. Permission to calculate a proposed hedge does not imply permission to execute it. Fine-grained delegation may appear cumbersome, but it allows institutions to expand productive automation without surrendering control over unrelated actions.

The conceptual discipline is simple: never infer authority from capability. Authority is granted, bounded and reviewable. Once this principle is accepted, autonomy becomes much easier to discuss precisely.

2 Autonomy as delegated discretion

MAIN IDEA

Autonomy is a bounded grant of discretion over objectives, information, actions, resources, time and escalation—not an intrinsic model property.

If autonomy is not intelligence, what is it? The most useful answer for institutional finance is delegated discretion. A system is autonomous to the extent that it can choose and pursue permitted steps without asking a human to approve every intermediate decision.

The word *discretion* is important. Automation alone does not imply much discretion. A fixed program may execute thousands of transactions automatically according to a rule that leaves no meaningful choice. Conversely, an agent may exercise substantial discretion while researching a difficult question, deciding which documents to inspect and which hypotheses to test, yet still have no authority to change an external record. Automation concerns whether a process proceeds without manual intervention. Autonomy concerns the choices delegated within that process.

Delegation therefore has several dimensions. First is the objective: what outcome is the system being asked to pursue? Second is the action set: what operations may it perform? Third is information scope: what may it read, and what may it remember? Fourth is time: for how long does the delegation remain valid? Fifth is resource scope: what computational, financial or operational budget may it consume? Finally, there are escalation conditions: which observations or uncertainties require the system to stop and seek help?

Imagine an agent monitoring a corporate bond portfolio. The objective may be to identify material deterioration in issuer credit quality. The agent may read approved public filings and internal research, calculate specified ratios and prepare alerts. It may not communicate externally, alter portfolio positions or change the definition of materiality. It may operate continuously but must escalate when two authoritative sources disagree, when a required document is unavailable or when an issuer enters a category not covered by the existing methodology.

This is a meaningful form of autonomy even though the agent cannot trade. It has discretion over how to pursue the monitoring assignment within a bounded domain. The boundaries are what make the delegation intelligible.

Human delegation offers a useful analogy. A portfolio manager does not tell an analyst every page to read. The analyst is trusted to pursue the assignment, but the trust is contextual. The analyst may not commit the fund to a transaction. A trader may have authority within position and risk limits but not authority to change those limits. A chief executive may possess broad discretion but remains constrained by law, the board and the company's governing documents. Autonomy has always existed inside institutions as bounded discretion.

AI makes the boundaries more programmable. That is both an opportunity and a responsibility. Permissions can be encoded, budgets enforced, actions logged and stopping conditions monitored. But poorly designed boundaries can also scale errors. A vague human instruction may be corrected through social context; a vague machine delegation can be repeated thousands of times. Precision therefore becomes part of governance.

3 A ladder of bounded delegation

MAIN IDEA

Delegation can progress from information and advice to approved action, bounded execution and controlled adaptation without assuming that more is always better.

It is helpful to describe autonomy as a ladder, provided we resist the temptation to treat the ladder as a race toward the top. The purpose of the ladder is to distinguish forms of delegation, not to rank institutions by technological ambition.

At the first level, the system advises. It produces analysis, drafts or recommendations for human use. The human remains the actor. This level can still be extraordinarily valuable. A system that reads thousands of documents and prepares traceable analysis may transform productivity without possessing authority to change anything outside

its workspace.

At the second level, the system prepares an action and executes only after specific approval. This is more than advice because the proposed action is operationally complete. A treasury agent might prepare a transfer, a trading agent might prepare an order, or a compliance agent might prepare a filing. The human approval must attach to the actual action and version, not to a vague category of future behavior.

At the third level, the system performs bounded execution. Certain action classes are authorized in advance within explicit limits. A service may send internal alerts when defined evidence conditions are met. An operations agent may reconcile routine exceptions below a threshold. A trading system may execute within a narrow mandate, subject to pre-trade controls and continuous monitoring. Here the institution has delegated genuine operational discretion.

A fourth level introduces controlled adaptation. The system may propose, or in carefully specified cases make, changes to parts of its own procedure. This requires a new distinction: authority to act under a rule is not authority to change the rule. An agent authorized to classify cases using a threshold should not automatically be authorized to move the threshold. An execution system allowed to route orders should not infer authority to rewrite its risk limits.

This is where discussions of self-improving AI can become conceptually dangerous. Improvement sounds universally desirable, but institutional procedures embody objectives, risk tolerances and legal constraints. A system that changes them is participating in governance, not merely optimizing performance. Controlled adaptation therefore requires independent evaluation, explicit change authority and the ability to restore a prior approved state.

The same service can occupy different levels simultaneously for different actions. A research agent may autonomously retrieve documents, automatically update a provisional working file, require approval before changing the approved investment thesis and have no authority whatsoever to trade. Describing the service as simply “autonomous” would conceal precisely the distinctions that governance needs.

The ladder therefore encourages a better managerial conversation. Instead of asking whether the firm should adopt autonomous AI, we ask which responsibilities are sufficiently specified, observable and reversible to justify which degree of delegation. The answer can differ across activities and can evolve as evidence accumulates.

4 Governance as architecture rather than an afterthought

MAIN IDEA

Permissions, identity, evidence, monitoring, cybersecurity, recovery and revocation must be built into the action path rather than added after deployment.

Once authority is delegated, governance cannot be reduced to a policy document or a final human check. It must become part of the architecture through which the system operates.

Begin with identity and permissions. The system should act through credentials that reflect its actual role rather than borrowing the unrestricted access of a developer or senior employee. If the research agent only needs to read approved filings, its technical permissions should express that fact. If it may create a draft but not publish it, the distinction should exist in the underlying systems, not merely in a sentence inside a prompt.

Evidence is another architectural requirement. A consequential conclusion should carry enough provenance that a reviewer can understand what supports it. In the research service, an alert about deteriorating liquidity should identify the observations, source versions, calculations and assumptions that produced the conclusion. The objective is not to generate an ornate explanation after the fact. It is to preserve an evidence chain as the work occurs.

Monitoring is equally important because delegated systems operate over time. A model that passed an evaluation in September may encounter a different distribution of cases in December. A data source may change format. A tool may begin returning incomplete responses. A memory store may accumulate stale assumptions. Governance therefore needs signals about the continuing health of the process: coverage, error rates, unresolved exceptions, action frequency, unusual tool use and deviations from expected behavior.

Escalation converts monitoring into institutional control. A system should know when it is outside its competence or authority. Missing evidence, conflicting sources, an unusual transaction size or an unexpected response from a tool may all justify stopping. The ability to stop is not evidence of weakness. It is evidence that the architecture recognizes its boundary.

Recovery matters because actions fail. If an agent sends a request and receives no confirmation, should it retry? If it retries, could the action be duplicated? If a workflow is interrupted halfway through, what state is authoritative? Financial operations have long developed concepts such as reconciliation, idempotency and rollback because repeated or partial actions can have real consequences. Agentic systems inherit these

old operational problems in a new interface.

Cybersecurity belongs inside this architecture from the beginning. Connecting models to tools, memory and other agents expands the attack surface. An untrusted document can attempt to manipulate instructions. A compromised credential can transform an analytical assistant into an unauthorized actor. Sensitive information can move across boundaries through seemingly innocent prompts or tool calls. Security is therefore not a later stage reached after the AI becomes sophisticated; it is a condition for granting access in the first place.

Finally, governance requires withdrawal. Every delegation should have a mechanism by which authority can be suspended. A human owner should know how to stop the service, revoke credentials, freeze memory writes or return to a simpler baseline. A system that can be authorized but cannot be reliably de-authorized is not well governed.

These controls may sound technical, but their purpose is deeply institutional. They ensure that the formal allocation of authority is reflected in the actual behavior of the system.

5 The objective is governable autonomy

MAIN IDEA

The aim is to expand discretion when evidence supports it and contract discretion when conditions, controls or operating assumptions deteriorate.

The phrase *governable autonomy* captures the destination of Lecture 2. The objective is not to keep humans manually involved in every step forever, nor is it to remove them as quickly as possible. The objective is to create a system in which discretion can expand when the evidence justifies expansion and contract when conditions no longer support it.

This changes the economics of adoption. A firm does not need to solve the problem of “full autonomy” before receiving value. It can begin with advice, where the system improves analysis but has little operational reach. It can then permit approved actions, reducing friction while preserving case-specific human authorization. Certain routine activities may graduate to bounded execution after testing demonstrates that controls are effective. Adaptation can remain under a separate and more demanding governance process.

The progression should be evidence-based. What would justify moving an action from advice to approved execution? Perhaps the institution has measured accuracy,

reviewed failure cases, established reliable provenance and demonstrated that the proposed action is correctly constructed. What would justify bounded execution? In addition to analytical quality, the institution may need enforced limits, monitoring, tested recovery and confidence that exceptional cases are detected. Each increase in discretion asks a different evidentiary question.

Reversibility should influence the decision. An internal draft can usually be corrected at low cost. A client communication is harder to retrieve. A market order may be impossible to reverse at the original price. A regulatory filing may create legal obligations. The less reversible the consequence, the stronger the case for explicit approval, narrower limits or additional independent controls.

Frequency also matters. Human review that works for ten cases a week may become superficial at ten thousand cases a day. At scale, nominal human oversight can become a fiction if the reviewer cannot meaningfully inspect the work. Governable autonomy therefore requires honest design. Sometimes stronger automated controls and selective escalation are safer than pretending that a person reviews every action.

The human role changes accordingly. Humans become designers of mandates, owners of exceptions, reviewers of evidence, authorities for consequential changes and governors of the system's evolution. This is not a retreat from responsibility. It is a movement toward higher-level responsibility. The institution must decide which judgments should remain human precisely because those judgments express objectives, values, fiduciary duties and risk appetite.

A board can use the same framework. Rather than asking "Are we ready for autonomous AI?" it can ask: Which action classes are delegated today? What limits apply? What evidence supported that delegation? Which controls are technically enforced? What happens when the system encounters an unknown case? Who can suspend it? Which parts of the system may change, and who authorizes those changes? These questions are concrete enough to govern.

Governable autonomy is therefore neither technological maximalism nor defensive minimalism. It is institutional craftsmanship. It recognizes that capable systems can create enormous value when they are allowed to do useful work, while insisting that authority remains a deliberate design choice.

A practical thought experiment: the morning after delegation

MAIN IDEA

A delegation becomes real when the system encounters missing evidence, novel cases and routine actions on the first day of authorized operation.

One final thought experiment makes the framework concrete. Imagine that the investment committee has approved a limited delegation on Monday evening. Beginning Tuesday morning, the research service may send internal credit alerts without waiting for an analyst, provided that the evidence comes from approved sources, two specified liquidity tests are satisfied and the alert contains the underlying calculations. Nothing dramatic has changed in the model. Yet the institution wakes up on Tuesday with a different operating system.

At 8:15 a filing arrives. The service recognizes the issuer, retrieves the approved prior state and calculates the new metrics. One source is delayed, but the remaining evidence is sufficient for one ratio and insufficient for another. This is the moment at which the quality of the delegation becomes visible. A poorly designed system may fill the missing field, infer that the conditions are met and send the alert. A governed system represents the missing evidence explicitly and asks whether the mandate permits action under partial information. If it does not, the service escalates. The most important behavior may be the decision not to act.

At 10:30 another issuer triggers the conditions cleanly. The alert is sent, its evidence is preserved and the portfolio manager acknowledges it. Governance has not obstructed autonomy; governance has made useful autonomy possible. The manager did not approve the individual alert because the institution had already approved the class of action and the conditions under which it could occur.

At 14:00 a third case produces an unusual pattern that was absent from testing. The model is confident, but the system recognizes that the issuer falls outside the categories covered by the mandate. Again it stops. Confidence is not substituted for authority. The exception becomes information for the humans who govern the service. They may later decide that the new category can be incorporated, but that change is a separate decision.

By the end of the day the institution has learned something important. Autonomy did not mean removing people from the process. It changed where people entered the process. Instead of approving every routine step, they designed the mandate, examined exceptions, reviewed performance and retained authority over changes to the rules. Human attention moved from repetition toward governance.

This is the practical promise of the architecture developed in Lecture 2. When responsibility and authority are made explicit, automation can expand without becoming institutional ambiguity. The system can move quickly where the institution understands the problem and slow down where uncertainty exceeds the mandate. That is a far more useful ambition than autonomy for its own sake.

6 The delegation charter

MAIN IDEA

A delegation charter makes the objective, action space, limits, information rights, duration, evidence record and revocation owner explicit.

Delegation becomes governable when it is written as a charter rather than implied by access to a tool. The charter states the objective the system may pursue, the action classes it may take, the information it may use, the resources it may consume, the period for which the authority is valid and the conditions that require escalation. It also names the person or body that grants, monitors and can revoke the delegation.

The objective must be narrow enough to constrain behavior. “Improve portfolio performance” is not an adequate mandate because it leaves risk, horizon, liquidity and client obligations unspecified. “Prepare and prioritize internal credit alerts for securities already held, using approved issuer evidence and the current watch-list policy” is more governable. It does not answer every design question, but it rules out many unauthorized interpretations of success.

Action classes should be enumerated. Reading, calculating, drafting, updating an internal case, sending a message, submitting an order and changing a policy are different actions even if one model can technically perform all of them. Limits attach to the action class: amount, frequency, counterparty, jurisdiction, channel, data sensitivity and reversibility. The charter should state whether the system may combine actions or only execute one approved step at a time.

Information authority is equally important. Access to a source for one purpose does not imply permission to copy it into memory, use it for evaluation or expose it to every agent in a workflow. The charter specifies approved sources, user context, retention and restrictions on derived information. Resource authority sets budgets for compute, model calls, data services and human escalation, preventing an agent from converting open-ended planning into open-ended cost.

Temporal authority gives delegation a beginning, an expiry and review triggers. A mandate that was reasonable during ordinary markets may be inappropriate after

a major disruption. Material changes in models, data, tools, policy or operating conditions can suspend the charter pending review. Expiry forces the owner to decide affirmatively whether evidence still supports the delegation.

The charter finally defines the evidence record. Every consequential action should be associated with the applicable mandate version, observed state, supporting evidence, model and skill versions, tool response, approvals and reconciliation outcome. The purpose is not to create maximal paperwork. It is to make the delegation real enough that the institution can determine whether the system acted within it.

7 Evidence gates on the autonomy ladder

MAIN IDEA

Every increase in autonomous authority should pass an evidence gate matched to the specific function and consequence being delegated.

The autonomy ladder becomes practical when each transition is treated as an evidence gate. Moving from analysis to recommendation requires evidence that the system identifies relevant facts and communicates uncertainty. Moving from recommendation to approved execution requires evidence that the proposed action is correctly constructed and can be inspected efficiently. Moving to bounded execution requires enforced limits, reliable exception detection, reconciliation and tested recovery.

Parasuraman, Sheridan and Wickens proposed levels and types of automation that separate information acquisition, analysis, decision selection and action implementation [1]. Their central insight remains useful: automation is multidimensional. A system can have high autonomy in collecting information and low autonomy in implementing action. The institution should select a level for each function rather than attach one autonomy label to the whole service.

Evidence should match the consequence. For a drafting assistant, evaluation may focus on source faithfulness, confidentiality and reviewer effort. For a payment service, the institution also needs transaction integrity, beneficiary controls, amount limits, duplicate prevention, sanctions checks, authorization and recovery. A high score on language tasks does not provide evidence for those operational responsibilities.

The gate should include normal cases, difficult cases and boundary cases that the system must refuse. Testing only successful execution rewards action without measuring restraint. Unknown entities, stale prices, conflicting approvals, closed markets, missing legal terms and tool timeouts reveal whether the system respects the mandate when completion is inconvenient.

Promotion should be reversible. A pilot can begin with small amounts, a restricted population and enhanced monitoring. If correction rates, control failures or environmental changes exceed thresholds, authority contracts automatically or by owner decision. Demotion is not necessarily evidence that the project failed. It is evidence that the governance system can respond to a changed risk estimate.

The gate also tests the surrounding organization. Are escalations answered within the required time? Can humans understand the evidence? Does nominal review become a bottleneck? Are controls independent of the model's own confidence? Autonomy is a property of the complete delegation arrangement, so the evidence must include people, process and technology.

8 From written policy to executable constraint

MAIN IDEA

Where institutional limits can be represented objectively, an independent action gateway should enforce them at the moment of execution.

A policy that exists only in prose may fail at machine speed. Wherever a constraint is objective enough to encode, the system should enforce it in the action path. Permitted instruments, maximum amount, approved recipients, trading hours, data classifications and required approvals can often be represented as structured rules evaluated immediately before execution.

This does not eliminate interpretation. Policies contain ambiguous terms, exceptions and relationships that depend on context. The architecture should distinguish the machine-enforceable core from cases requiring judgment. A rule can block an order above a limit. A human may need to determine whether several related orders constitute one economic exposure or whether an emergency exception is justified.

The enforcement point should be outside the generative model. The model may propose an action and explain why it believes the action is permitted, but the tool gateway checks identity, role, mandate, parameters and current limits. This separation creates a control that fails differently. It also prevents prompt content from redefining authority.

NIST Special Publication 800-53 provides a broad catalog of security and privacy controls including access control, audit, configuration management and incident response [3]. An agentic system gives these familiar controls a new interface. The identity invoking a tool may represent a service acting on behalf of a user under a particular mandate. Authorization therefore needs to bind user, service, purpose and

action rather than rely on a shared technical account.

Policy versions must travel with actions. If a limit changes after a plan is generated, the execution gateway applies the current rule unless the institution has explicitly designed otherwise. Simulations and post-incident reviews may need the historical version to reproduce what was permitted at the time. Versioned policy is both an operating control and an evidentiary record.

Exceptions should not become hidden bypasses. A break-glass path needs stronger authentication, a reason, a narrow duration, enhanced logging and retrospective review. If emergency use becomes frequent, the normal policy or process may be poorly designed. Governable autonomy makes exception patterns visible to the owner rather than allowing informal workarounds to accumulate.

9 Meaningful human oversight

MAIN IDEA

Human oversight is meaningful only when reviewers have understandable evidence, sufficient time, genuine authority and a manageable volume of cases.

Human oversight is often invoked without specifying what the person can realistically observe or change. A reviewer who receives thousands of complex cases, seconds before execution, does not provide meaningful oversight merely because an approval button exists. The design should match the volume and consequence of work to human attention.

Meaningful review requires a comprehensible evidence package, sufficient time, authority to refuse, and protection against incentives that reward automatic approval. The reviewer should know which fields were retrieved, calculated or inferred; what changed from the prior state; what controls already ran; and why the case was escalated. Interfaces should direct attention to decision-relevant uncertainty rather than display all available text.

Automation bias can cause people to accept a system's recommendation, while algorithm aversion can cause them to reject useful automation after visible errors. The remedy is not to assume perfect neutrality. It is to design calibrated roles. Human review is most valuable where objectives conflict, novel context matters, evidence is incomplete or accountability requires an authorized judgment. Automated checks are often stronger for high-volume consistency, arithmetic and hard limits.

Endsley and Kiris showed that excessive automation can reduce situation awareness and make recovery difficult when a person must retake control [2]. A handoff is

therefore a performance condition, not a slogan. The person needs an intelligible state, a clear reason for the handoff and tools appropriate to the remaining time. If the system waits until it is confused at the final second, human presence may not prevent harm.

Oversight also occurs at different time scales. Case-level oversight approves or rejects an action. Operational oversight monitors queues, exceptions and service health. Governance oversight reviews patterns, changes mandates and allocates accountability. Board oversight challenges whether the portfolio of delegations remains compatible with risk appetite and strategy. Collapsing these roles into “human in the loop” hides the decisions that people actually need to make.

At scale, selective review can be more honest than universal review. Low-risk, well-observed cases may execute within limits, while unusual or high-impact cases escalate. Samples of routine cases can be audited to detect drift. The selection logic itself must be tested so that the service cannot make its own difficult cases invisible.

10 Revocation, containment and recovery

MAIN IDEA

Delegated authority is governable only if it can be withdrawn quickly, contained independently and restored through explicit evidence.

Authority is not governable unless it can be withdrawn. Revocation should be designed before delegation begins. The owner needs a direct mechanism to disable an action class, tool, model version, source or entire service. The mechanism should not depend on the component being disabled and should propagate quickly enough for the consequence involved.

Triggers can include control failure, abnormal correction rates, a cybersecurity incident, an unauthorized data flow, a material model change, market disruption or loss of an authoritative source. Some triggers can cause automatic contraction to a safer mode; others require a named decision-maker. The response may move the service from bounded execution to recommendation, from recommendation to draft, or from operation to complete suspension.

Containment preserves evidence and prevents expansion. Credentials can be revoked, queues paused, network paths restricted and mutable state protected. Recovery then determines which intended actions completed, which failed and which remain uncertain. Reconciliation is essential because stopping the service does not undo actions already taken.

Operational-resilience principles emphasize the ability to deliver critical operations through disruption and to learn from severe but plausible scenarios [4]. For an autonomous service, resilience includes a manual or simpler automated fallback, an up-to-date runbook, tested restoration and clear communication to users who might otherwise rely on stale outputs.

Restart criteria should be explicit. Fixing the visible defect may be insufficient if the institution does not understand the failure mechanism or cannot test the control. Restoration can be staged: read-only operation, shadow mode, limited pilot and then full delegated scope. The same evidence logic used for initial promotion applies after an incident, with additional attention to whether corrective action addresses the causal pathway.

Revocation is also relevant at ordinary lifecycle events. A vendor changes a model, an employee changes role, a strategy closes or a policy expires. Access and delegation should not persist through organizational inertia. Periodic recertification asks owners to confirm that the purpose, controls and economics still justify the authority.

11 Worked case: bounded treasury action

MAIN IDEA

Treasury automation is credible when it operates inside a small, observable action space with enforced limits, reconciliation and safe fallback.

Consider a treasury service that forecasts daily cash positions and prepares transfers among approved internal accounts. At the advisory level, it retrieves balances, known flows and settlement obligations; identifies a projected deficit; and recommends a transfer. A treasurer reviews the evidence and submits the action through an existing system.

The next delegation level allows the service to prepare a fully constructed transfer for approval. Evidence must show that account identity, amount, currency, value date and purpose are correct, and that duplicate and balance controls operate. The person sees the projected position before and after the transfer, the data freshness and any assumptions about expected receipts.

Bounded execution might later permit transfers below a threshold among a fixed set of accounts during defined hours. The action gateway verifies service identity, current mandate, account set, amount, daily aggregate, available balance and policy version. A unique identifier prevents duplicate execution. Confirmation from the payment system is reconciled to the intended transfer, and uncertainty routes to an exception

queue rather than a blind retry.

The service does not gain authority to add beneficiaries, change thresholds or alter the forecast objective. Those are changes to the delegation boundary. It also cannot use a high confidence score to override stale balance data. If a critical source is unavailable, authority contracts to advice or stops entirely according to the charter.

Performance is measured across the full outcome: forecast quality, avoided overdrafts, transfer cost, exceptions, human review time and control events. The institution compares these results with the prior process and tests severe but plausible cases such as delayed settlements, duplicated messages, a closed payment rail and conflicting approvals.

This case shows that autonomy can be valuable without being general. The service has discretion inside a small action space whose state, limits and outcomes are observable. The arrangement can save time and respond consistently while leaving changes to beneficiaries, policy and risk appetite with authorized people. Boundedness is not a temporary embarrassment on the way to unlimited autonomy. It is the design feature that makes delegation credible.

12 The board view: a portfolio of delegations

MAIN IDEA

Senior governance should view autonomous services as a portfolio of authorities whose shared dependencies and aggregate consequences require oversight.

Senior governance should treat autonomous services as a portfolio of delegated authorities. For each service, leaders should know the business purpose, owner, action classes, maximum consequence, evidence supporting the current level, critical dependencies, monitoring and revocation path. A portfolio view reveals concentration that individual project reviews miss.

Several services may depend on the same foundation model, identity provider, data vendor or tool gateway. Separate mandates can therefore share a common failure. Aggregate limits may matter even when each service operates within its own threshold. A vendor outage or policy error can affect multiple business lines simultaneously. The portfolio should identify these dependencies and test correlated disruption.

Risk-management guidance such as ISO 31000 emphasizes integration, structured assessment, monitoring and continual improvement [5]. Applied to agentic systems, that means delegation decisions should fit the firm's ordinary governance rather than exist in a technological enclave. Material authorities belong in risk appetite, operational

resilience, security, model risk, change management and business ownership.

The board does not need to approve every prompt or tool call. It should challenge whether management can explain where autonomous authority exists, whether assurance matches consequence, whether human oversight is credible, and whether the institution can contract authority under stress. A clear portfolio makes those questions answerable.

13 Delegation stress test: when confidence and authority diverge

MAIN IDEA

High model confidence never substitutes for a missing condition of authority; safe systems continue to help at a lower delegation level.

The stress test begins with an internal service authorized to send routine collateral-call notifications when exposure, agreement terms and recipient identity are all verified. The model has high confidence in a new case, but the agreement repository contains two possible amendments and the latest effective version is unclear.

Participants first decide whether confidence can compensate for missing authority conditions. It cannot: the mandate requires verified terms. The appropriate behavior may be to prepare a draft, preserve the conflicting documents and escalate to the agreement owner. The system remains useful while declining the delegated action.

The scenario then changes. The agreement is verified, but the notification amount exceeds the automated threshold by one percent. The team tests whether the gateway enforces the limit or whether the model can persuade a tool that the difference is immaterial. A sound architecture routes the case to approval. Materiality judgments do not belong to a model attempting to pass its own action constraint.

Next, volume rises sharply during market stress. Human reviewers cannot examine every escalation within the contractual window. The group must choose among temporarily increasing authority, narrowing eligible cases, adding reviewers or degrading to a simpler rule-based process. Each option changes risk. Nominal human oversight is not accepted if queue size makes review impossible.

Finally, the model provider announces an immediate version change. Although the tool permissions remain the same, the evidence supporting the existing delegation may no longer apply. Participants identify whether the charter requires suspension, shadow comparison or enhanced monitoring. They determine who can make the

decision and how completed actions remain attributable to the prior version.

The deliverable is a revised delegation charter with objective, action class, limits, information, time, escalation, revocation and evidence record. The exercise is successful when the service can continue to add value at a lower authority level whenever a condition for higher authority fails. Governable autonomy is demonstrated by controlled contraction as much as by successful action.

Closing synthesis: from delegated authority to adaptive institutions

MAIN IDEA

Responsible autonomy delegates only what can be defined, observed, governed and withdrawn while preserving accountability for objectives and change.

Lecture 2 began by asking how intelligence becomes organized. We distinguished the model from the system around it, then introduced tools, skills and memory as different extensions of capability. We defined the agent as a bounded unit of responsibility and treated multi-agent arrangements as organizational designs rather than collections of digital personalities. That gave us an artificial organization.

Part II then asked what authority such an organization should receive. The answer began with a separation: capability is not authority. Autonomy is delegated discretion, and delegation has dimensions—objective, action scope, information, time, resources and escalation. A ladder from advice through approved action and bounded execution to controlled adaptation helps us specify the degree of discretion without assuming that more is always better. Governance becomes architectural through permissions, evidence, monitoring, cybersecurity, recovery, escalation and withdrawal.

The resulting principle is simple enough to remember: delegate only what you can define, observe and govern. Yet this principle does not imply that the architecture must remain static. Once a governed system operates, it produces feedback. It encounters new cases. It discovers weaknesses. People revise procedures. Models improve. Agents may be permitted to propose changes to their own workflows. At that point a new question appears.

What happens when the organization itself begins to change?

That is the bridge to Lecture 3. The next stage of the course moves from organized and governed intelligence to adaptation, recursive improvement, endogenous organization and the larger institutional strategy required when intelligent systems become participants in their own evolution.

References

MAIN IDEA

These sources support the treatment of automation levels, human oversight, security controls, operational resilience and risk management.

References

- [1] Raja Parasuraman, Thomas B. Sheridan and Christopher D. Wickens. “A Model for Types and Levels of Human Interaction with Automation.” *IEEE Transactions on Systems, Man, and Cybernetics—Part A*, 30(3), 2000.
- [2] Mica R. Endsley and Esin O. Kiris. “The Out-of-the-Loop Performance Problem and Level of Control in Automation.” *Human Factors*, 37(2), 1995.
- [3] National Institute of Standards and Technology. *Security and Privacy Controls for Information Systems and Organizations, SP 800-53 Rev. 5*. NIST, 2020.
- [4] Basel Committee on Banking Supervision. *Principles for Operational Resilience*. Bank for International Settlements, 2021.
- [5] International Organization for Standardization. *ISO 31000:2018 Risk Management—Guidelines*. ISO, 2018.